MVRL LAB MEETING · 23 OCT 2026

Does Remote Sensing Beat the Trail Map?

Predicting hiking speed in regions a model has never seen

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CSE 5999 independent study · Mentors Danni Beaulieu and Tianyi Gao · Advisor Prof. Nathan Jacobs

Background: lidar contours of Mt Washington, NH (USGS 3DEP). Slides made with help from Claude (Anthropic).

GIS still times hikes with a 1993 formula

TOBLER'S HIKING FUNCTION (1993)

$$v = 6\, e^{-3.5\,|s + 0.05|} \text{ km/h}$$

s = slope. Fastest, 6 km/h, on a gentle downhill.

- **Fit to a textbook table**
  not to real hikes
- **Slope is its only input**
  rocks, roots and brush are ignored

CAN THE TRAIL MAP HELP?

Share of walked trail with OpenStreetMap's difficulty tag:

96%

Adirondacks

1.5%

Green Mountains

Often the map can't say how hard a trail is.

Goal: predict speed in regions the model never saw

WHAT'S BEEN DONE

Refit the slope curve
with fitness-app data (Campbell et al. 2019, 2022)

Add map or lidar inputs
Wood et al. 2023; STRIDE 2024; RasterNet (this lab, 2020)

Gap: always tested in the same area they were trained on.

OUR TEST

Train on 16 regions, predict the 17th.
Repeat for each region.



Four models, same learning, different inputs:

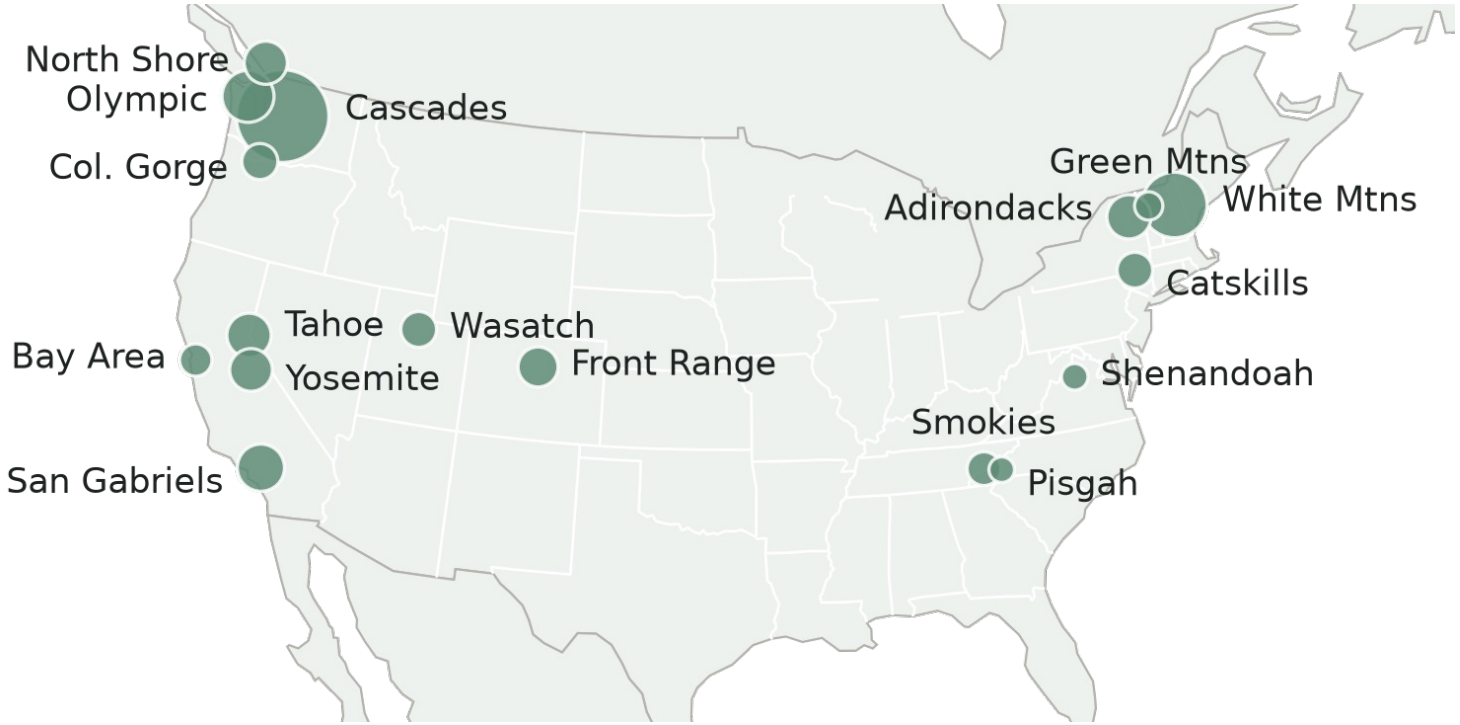
Slope only
like Tobler

Map tags
OSM trail tags

Remote sensing
lidar terrain, vegetation

Both
tags + remote sensing

Data: 1,992 real hikes in 17 regions



Circle size = hours of walking

1,992

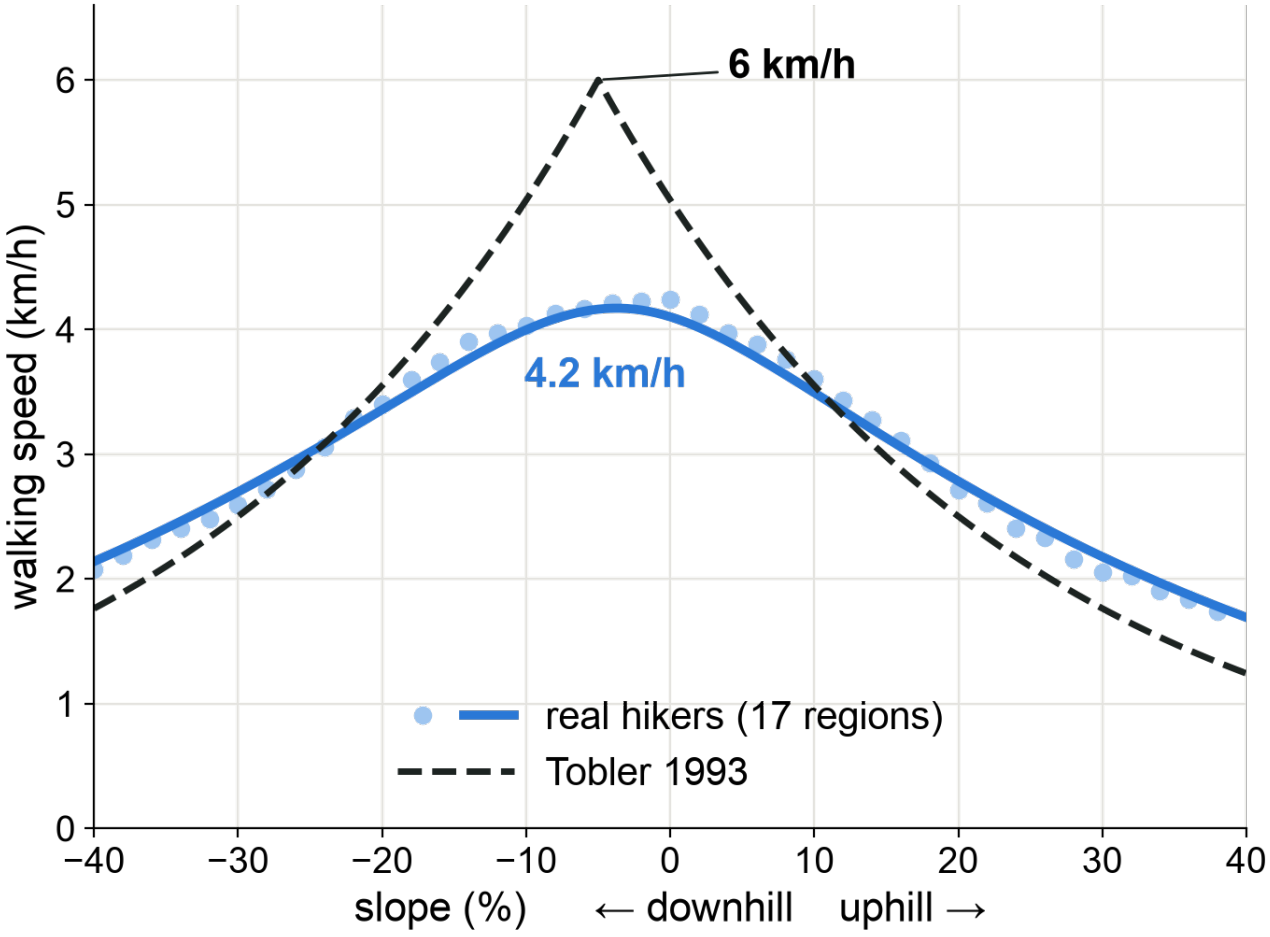
hikes, from public GPS traces

290,640

trail segments of 50 m, each with a measured speed

Public GPS traces from OpenStreetMap → Clean, remove stops, snap to the trail map → 50 m segments with slope, map tags and lidar

Tobler's formula is too fast for real hikers



4.2 km/h

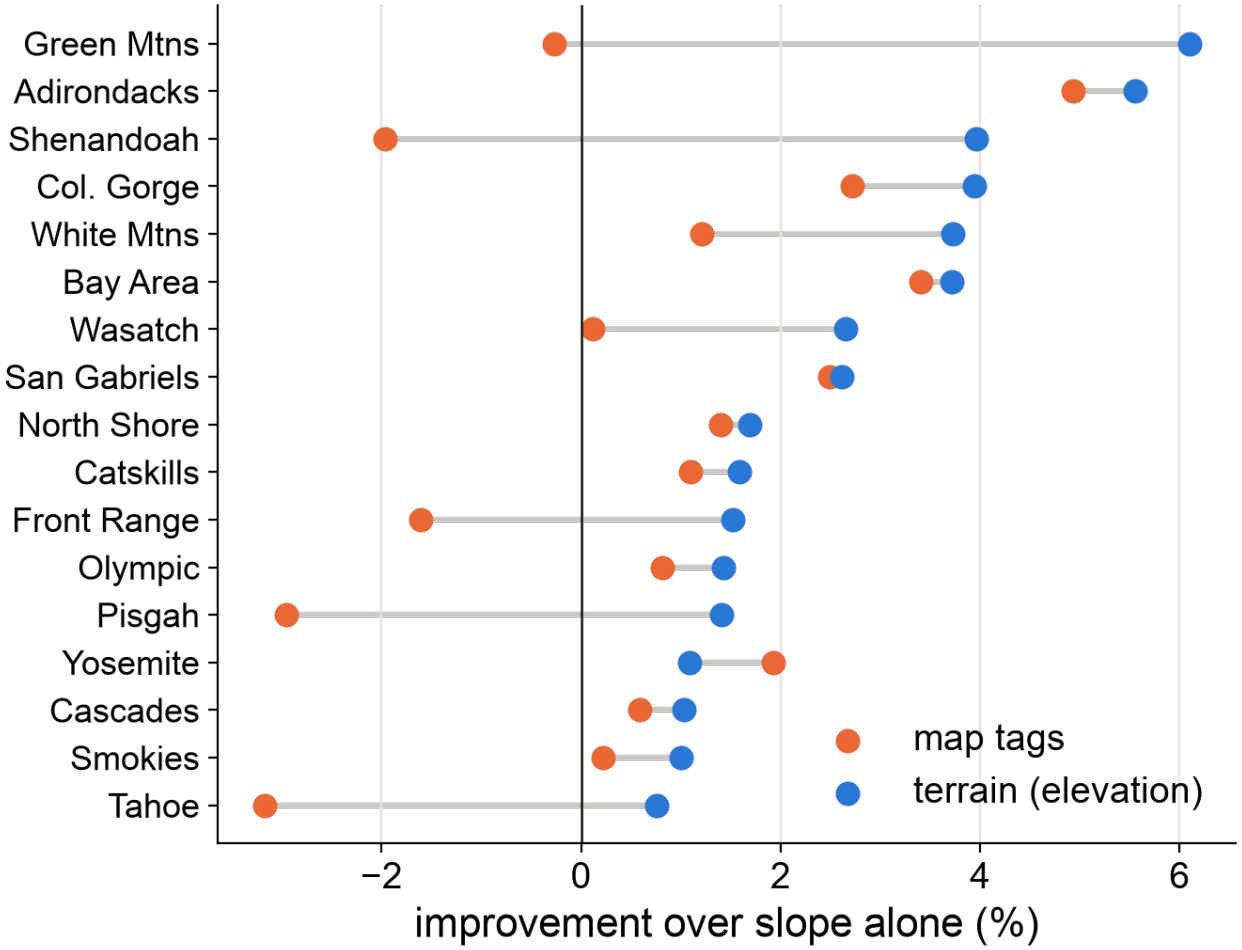
real hikers' top speed

Tobler says 6 km/h

17 of 17

regions where Tobler predicts worse than a curve fitted to real hikes

In new regions, terrain beats map tags



16 of 17

new regions where terrain predicts better than map tags

2–3%

terrain's gain over slope alone

map tags: under 1%

Each row: train on the other 16 regions, test on this one.

Next steps

Done so far: the dataset, the four-model comparison and a first paper draft.

Weeks 10–11	Learn Tobler's constants from terrain, with uncertainty
Week 12	Add aerial imagery (NAIP) with a pretrained encoder
January	Paper to IGARSS 2027

Question for you: which pretrained encoder would you use for 1 m aerial imagery along trails?